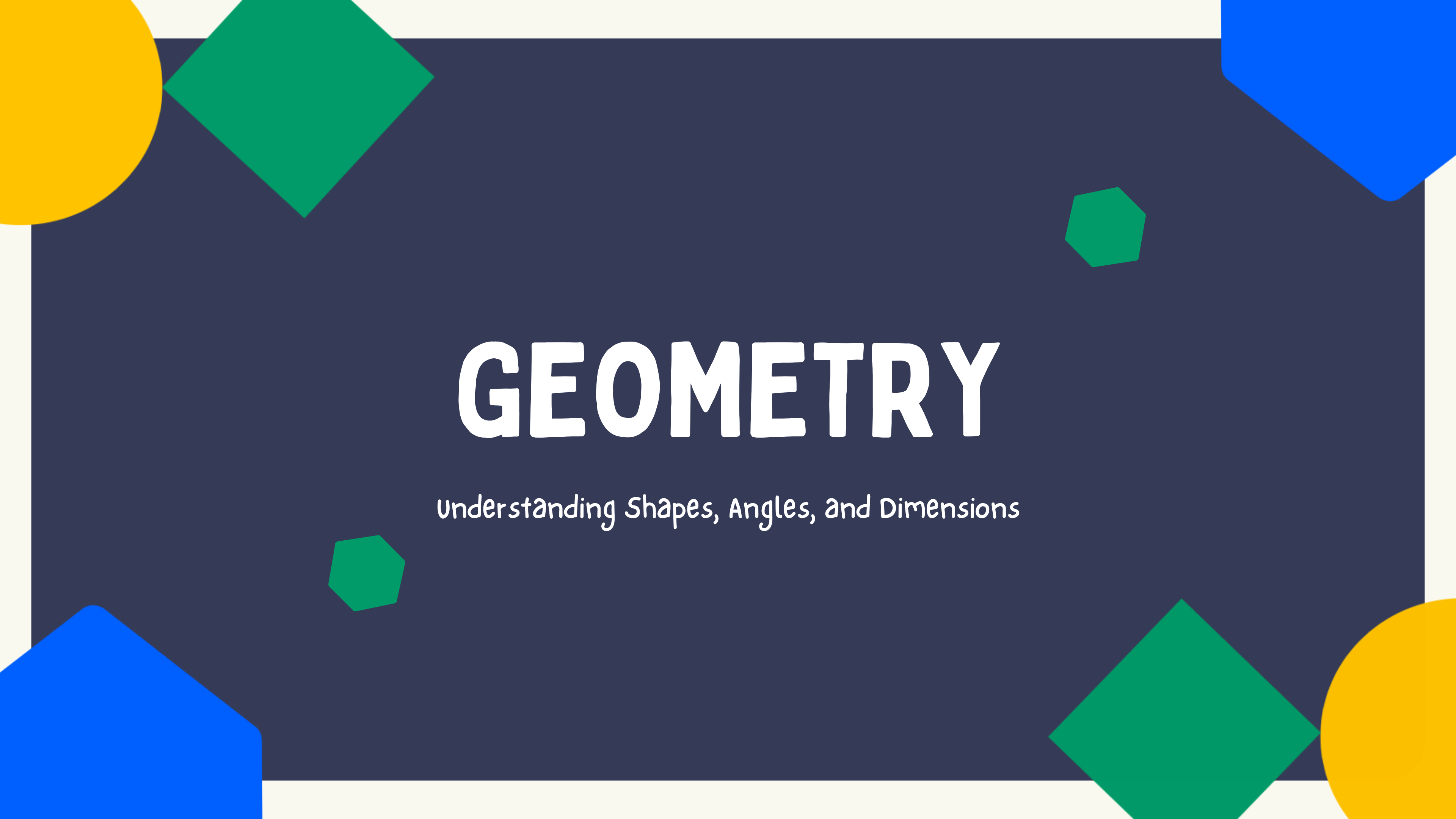


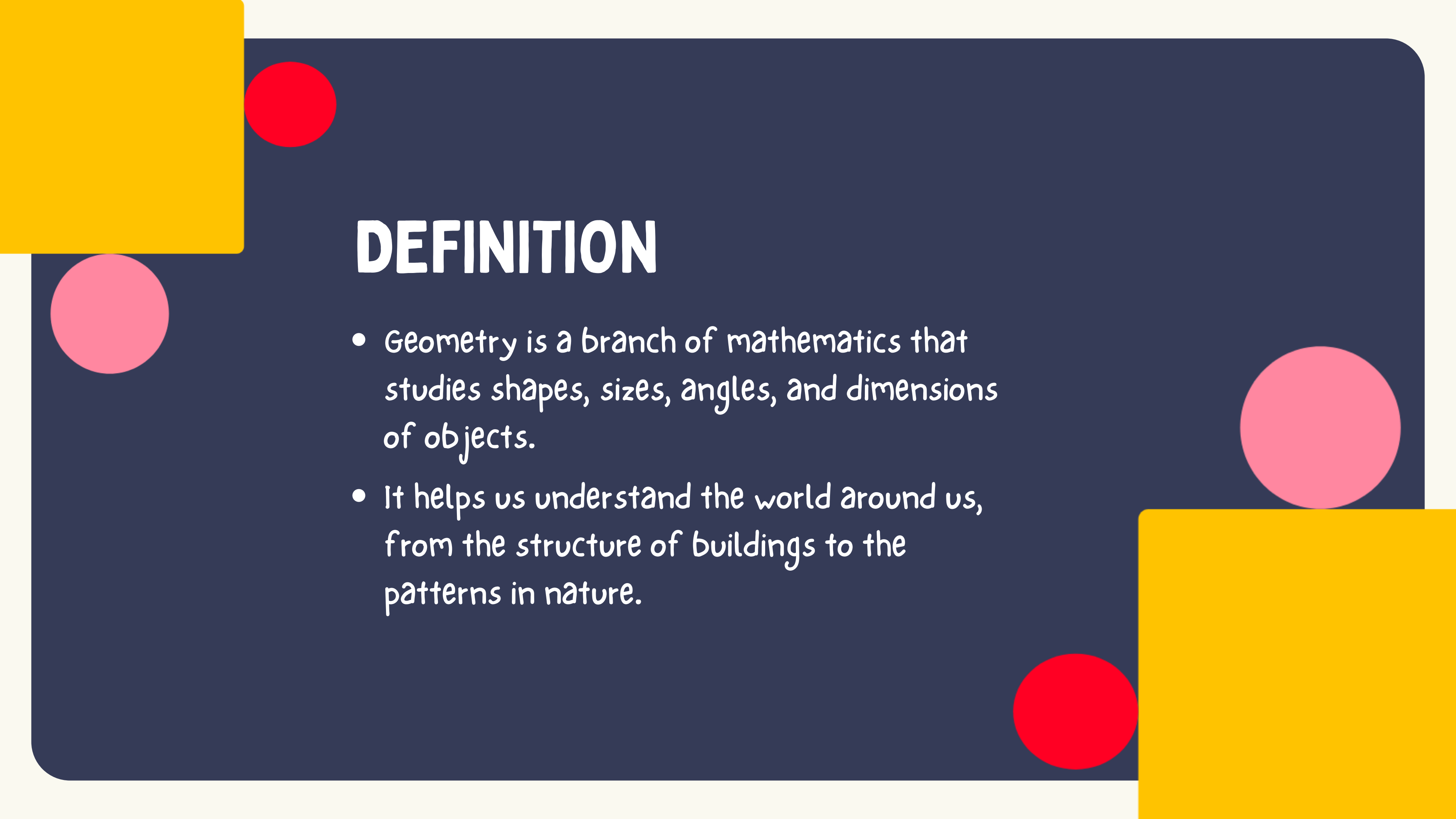


GEOMETRY

understanding Shapes, Angles, and Dimensions

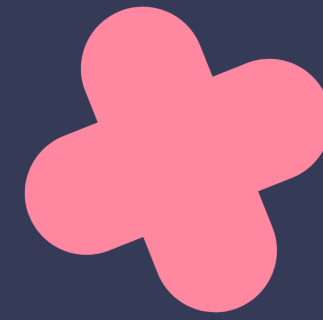
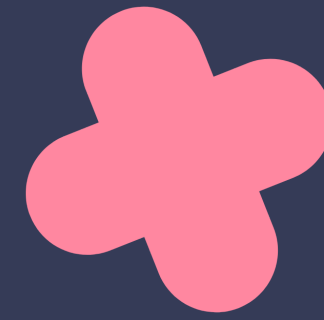
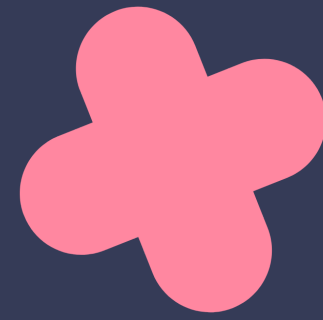
WHAT IS GEOMETRY?





DEFINITION

- Geometry is a branch of mathematics that studies shapes, sizes, angles, and dimensions of objects.
- It helps us understand the world around us, from the structure of buildings to the patterns in nature.

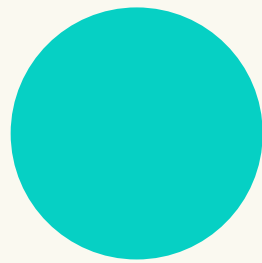


BASIC CONCEPTS IN GEOMETRY

Introduction to fundamental elements in geometry.



POINTS, LINES, AND PLANES



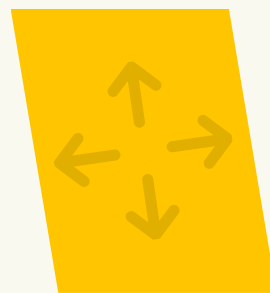
Point

A location in space with no size (denoted by a dot, like A).



Line

A straight path that extends infinitely in both directions.

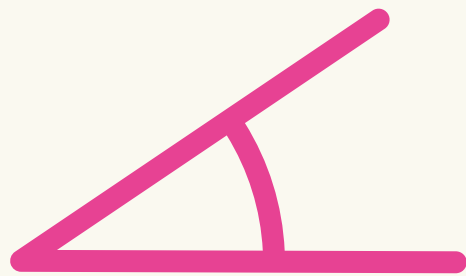


Plane

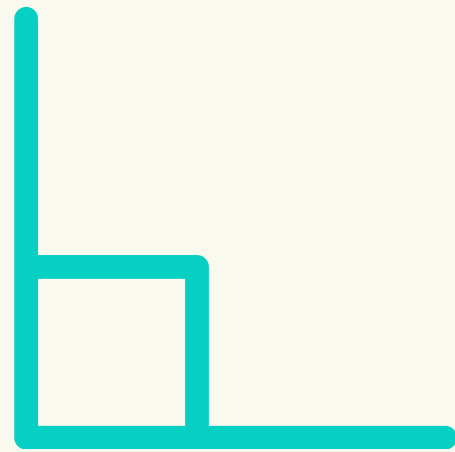
A flat surface that extends infinitely in all directions.

ANGLES

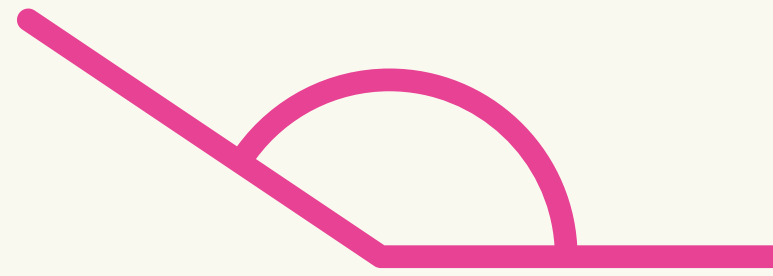
- Formed when two lines meet at a point.
- Types of angles:



Acute angle ($< 90^\circ$)



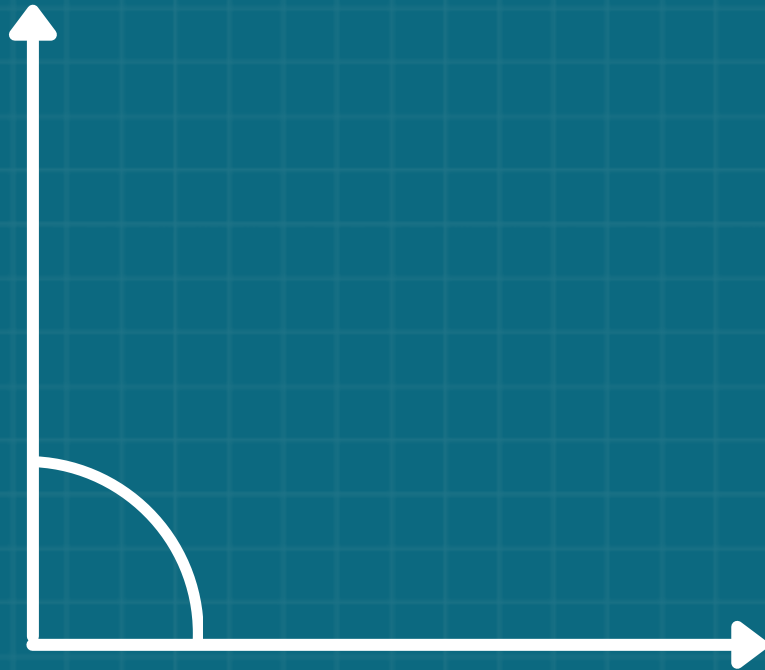
Right angle (90°)



Obtuse angle ($> 90^\circ$ and $< 180^\circ$)

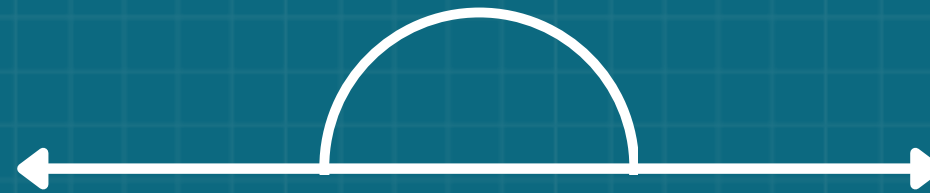


Straight angle (180°)



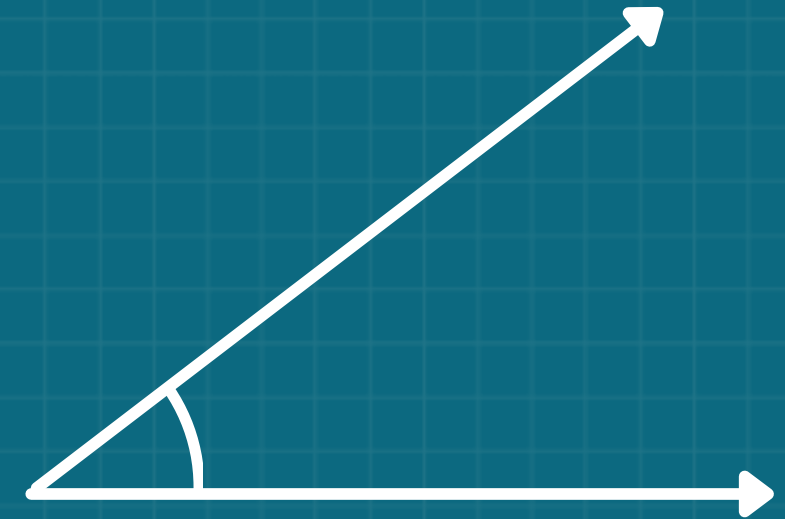
RIGHT ANGLE

It is an angle that
measures exactly 90
degrees



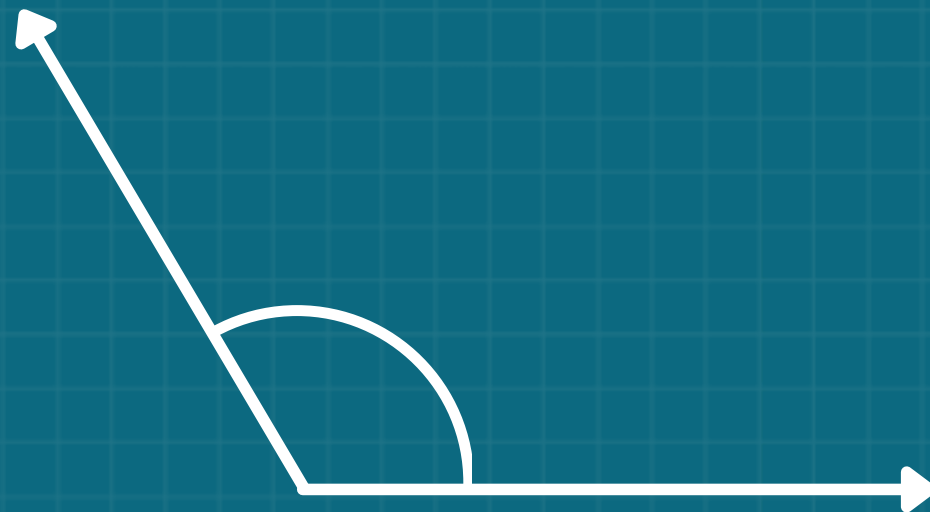
STRAIGHT ANGLE

It is an angle that
measures exactly 180
degrees



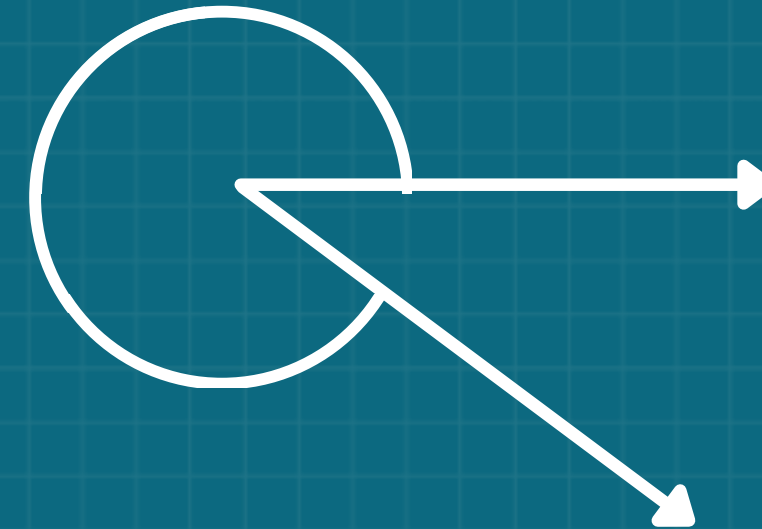
ACUTE ANGLE

It is an angle that
measures less than 90
degrees



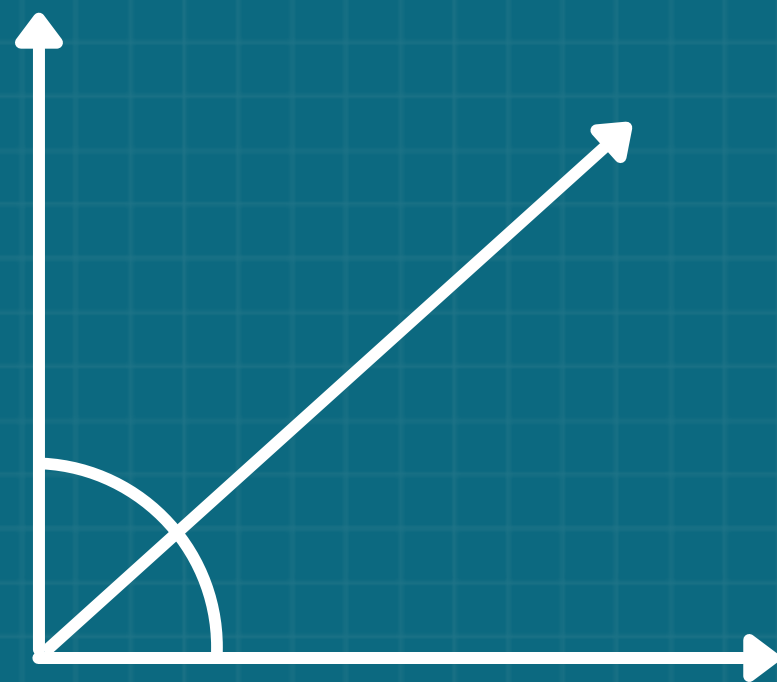
OBTUSE ANGLE

It is an angle that
measures between 90
and 180 degrees



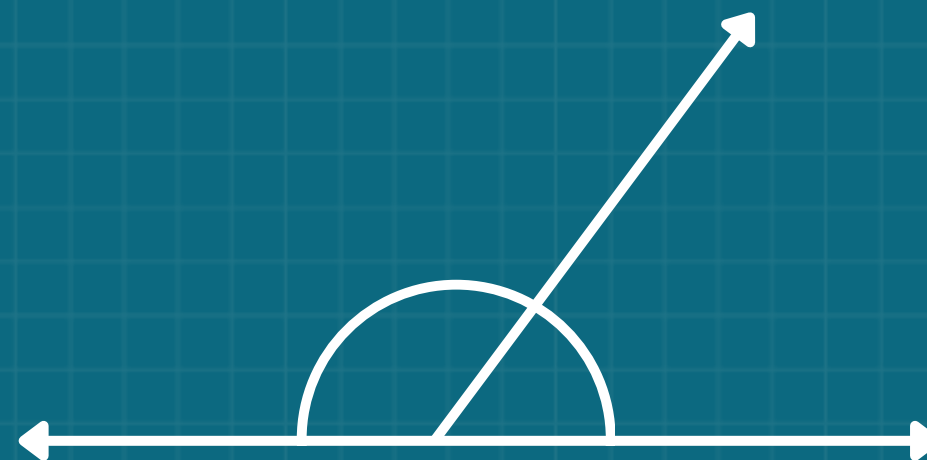
REFLEX ANGLE

It is an angle that
measures more than
 180° but less than 360°



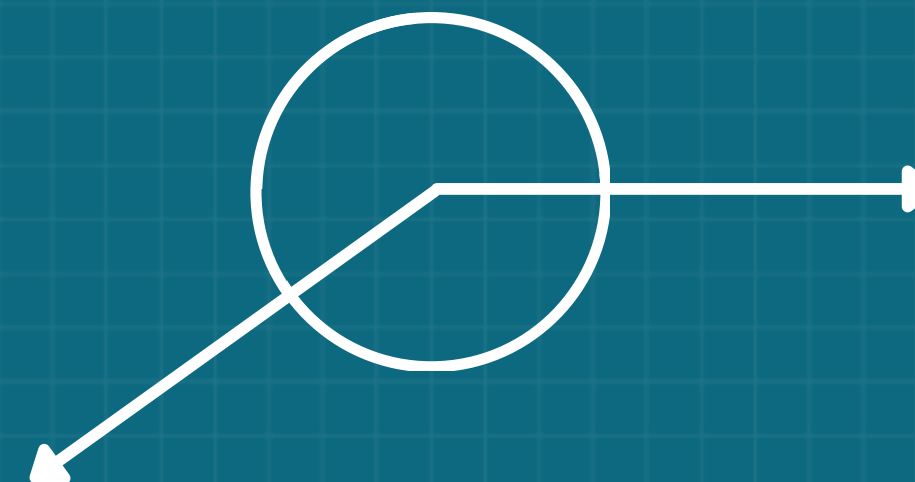
COMPLEMENTARY ANGLE

It is a pair of angles that when added are equal to 90°



SUPPLEMENTARY ANGLE

It is a pair of angles that when added are equal to 180°



CONJUGATE ANGLE

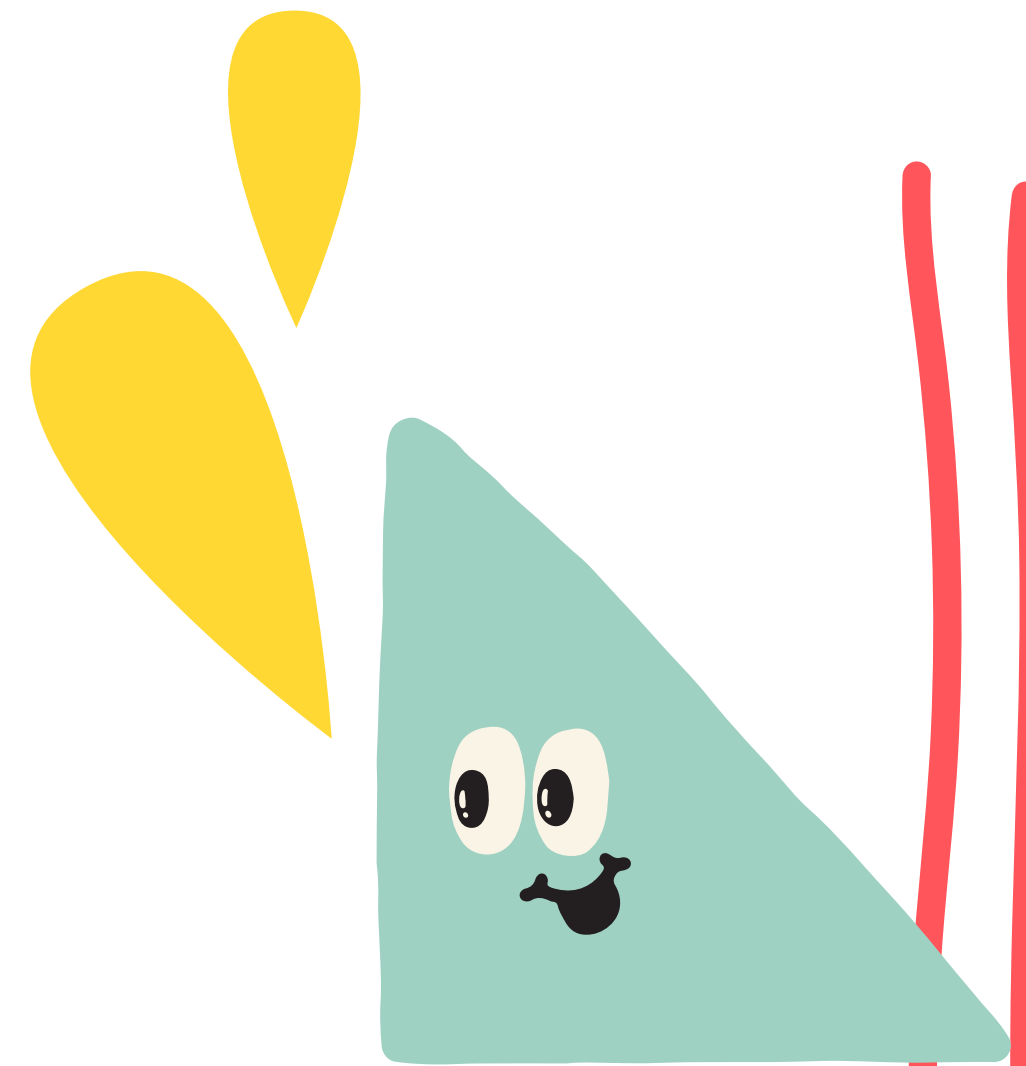
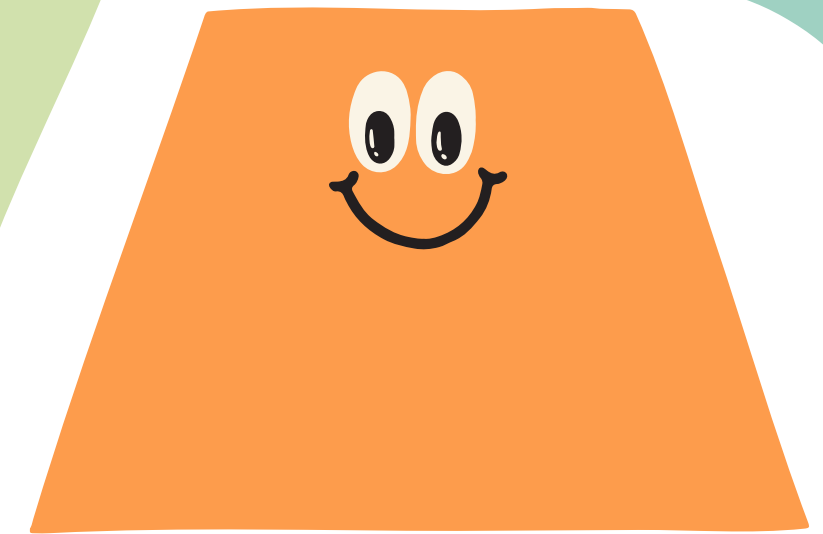
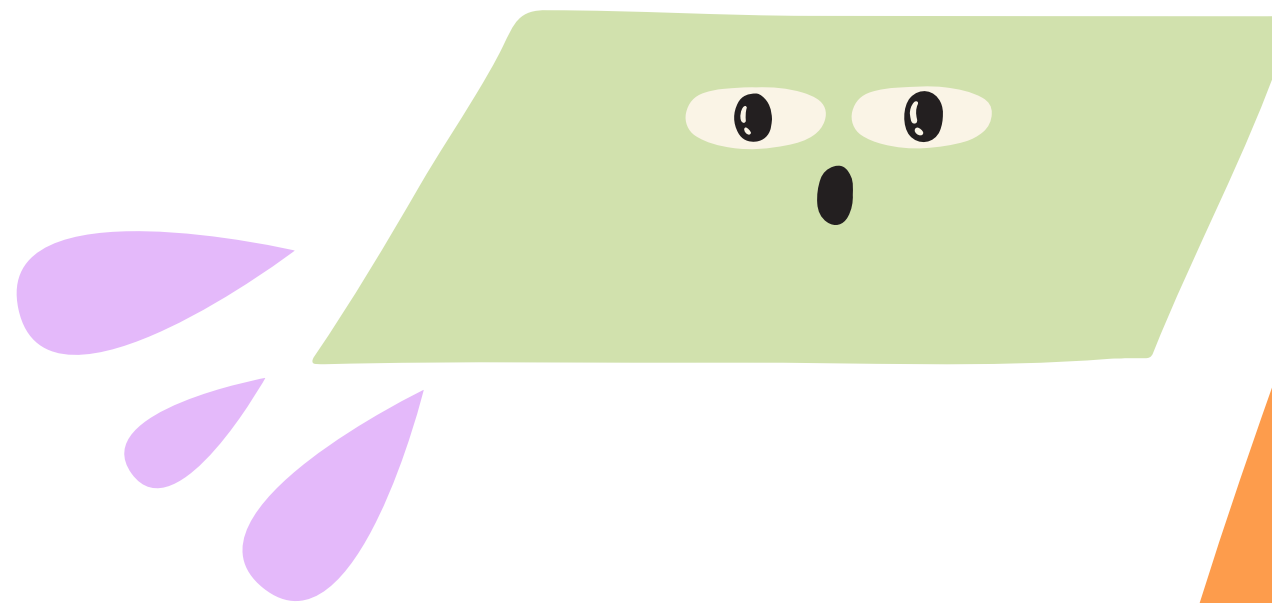
It is a pair of angles that when added are equal to 360°



AREA

WHAT IS IT?

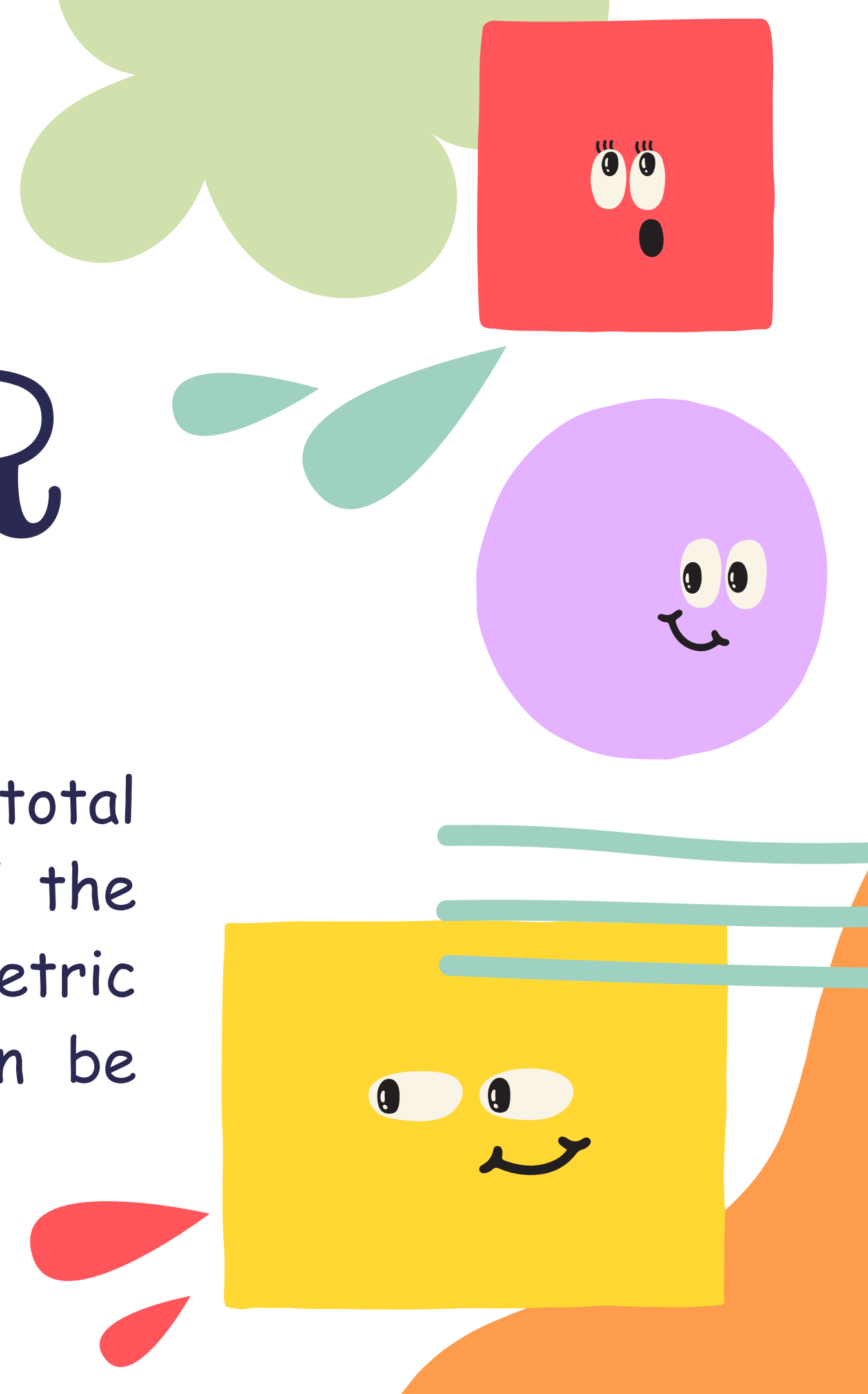
Area is the amount of space occupied by a two-dimensional figure. In other words, it is the quantity that measures the number of unit squares that cover the surface of a closed figure. The standard unit of area is square units.



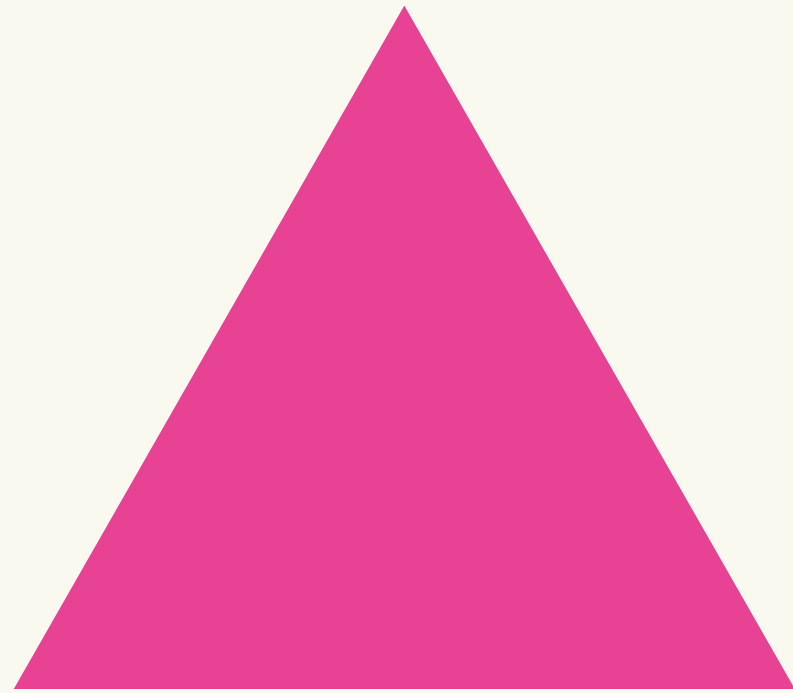
PERIMETER

WHAT IS IT?

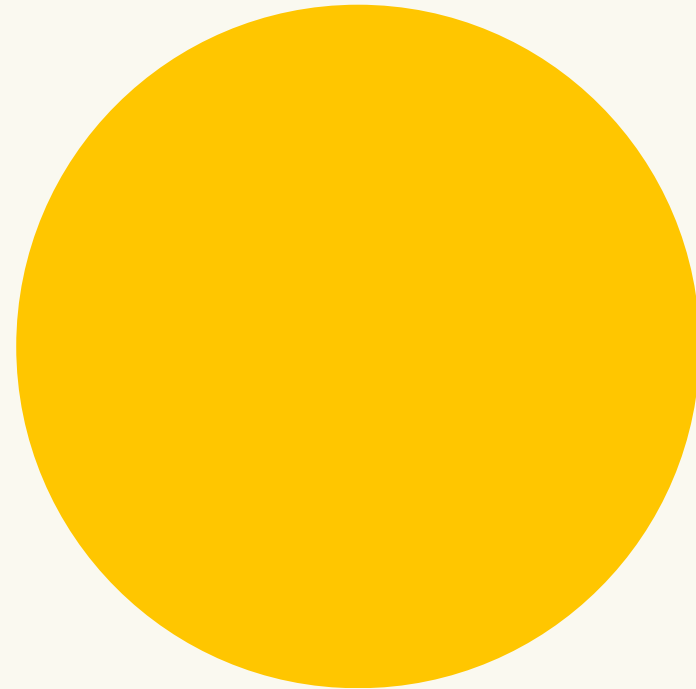
The perimeter of a shape is defined as the total distance around the shape. It is the length of the outline or boundary of any two-dimensional geometric shape. The perimeter of different figures can be equal in measure depending upon the dimensions.



2D SHAPES (FLAT SHAPES)



Triangle (3 sides)



Circle (A round shape with no sides)

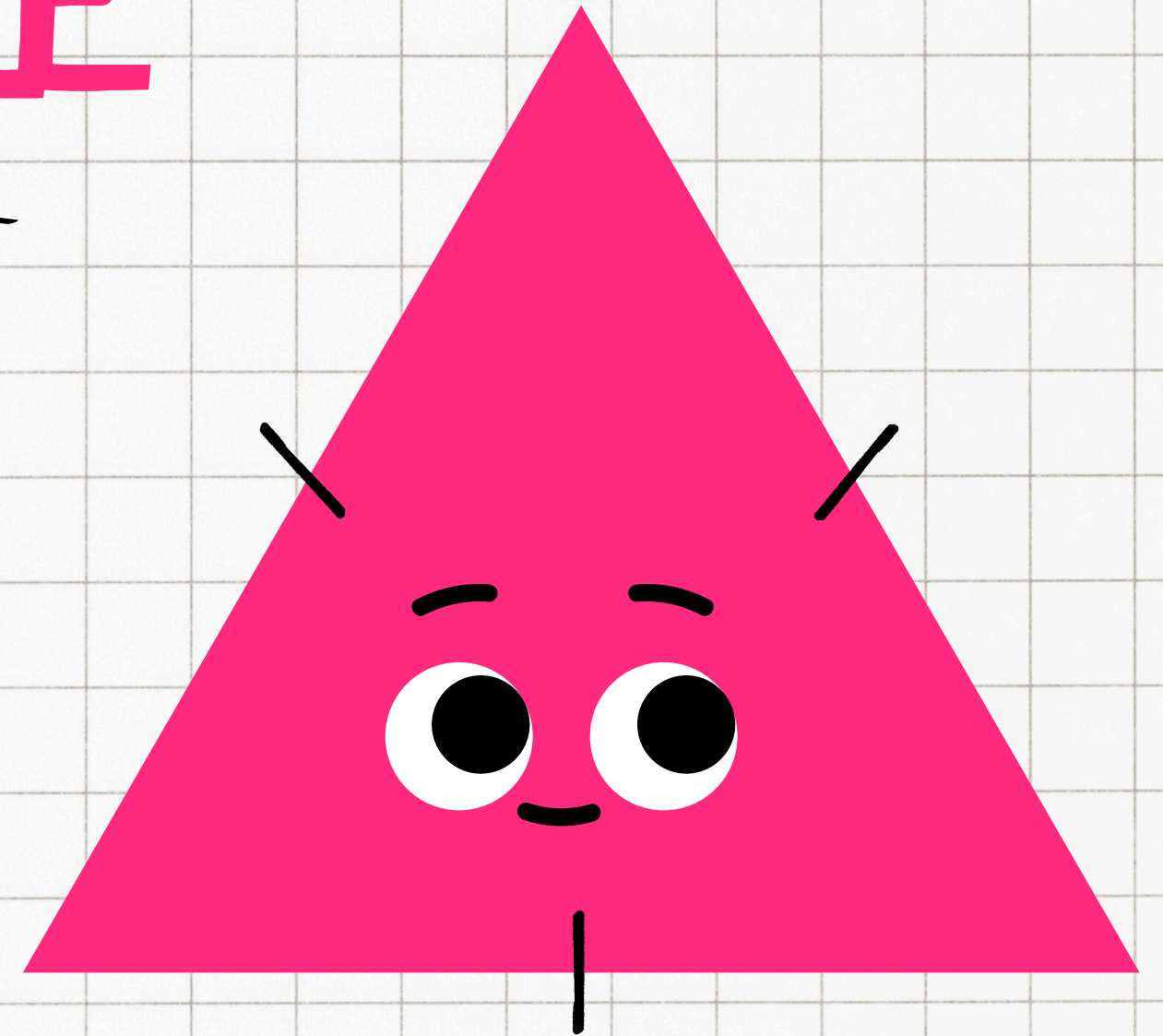


Quadrilateral (4 sides)

EQUILATERAL TRIANGLE

Definition: All three sides are of equal length.

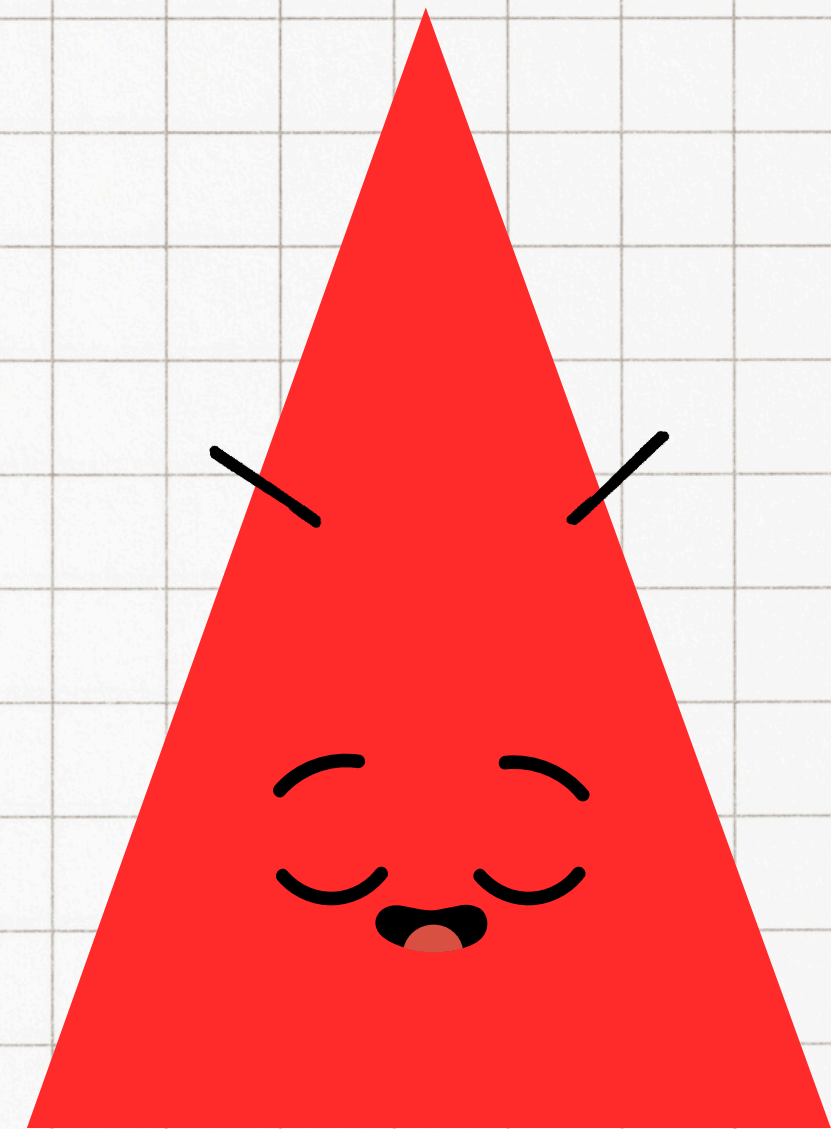
Properties: All three angles are equal, and each angle measures 60° .



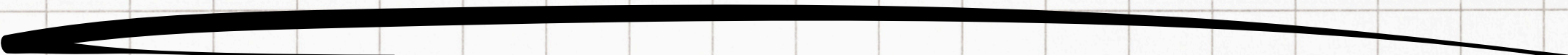
ISOSCELES TRIANGLE

Definition: Two sides are of equal length.

Properties: The angles opposite the equal sides are also equal.

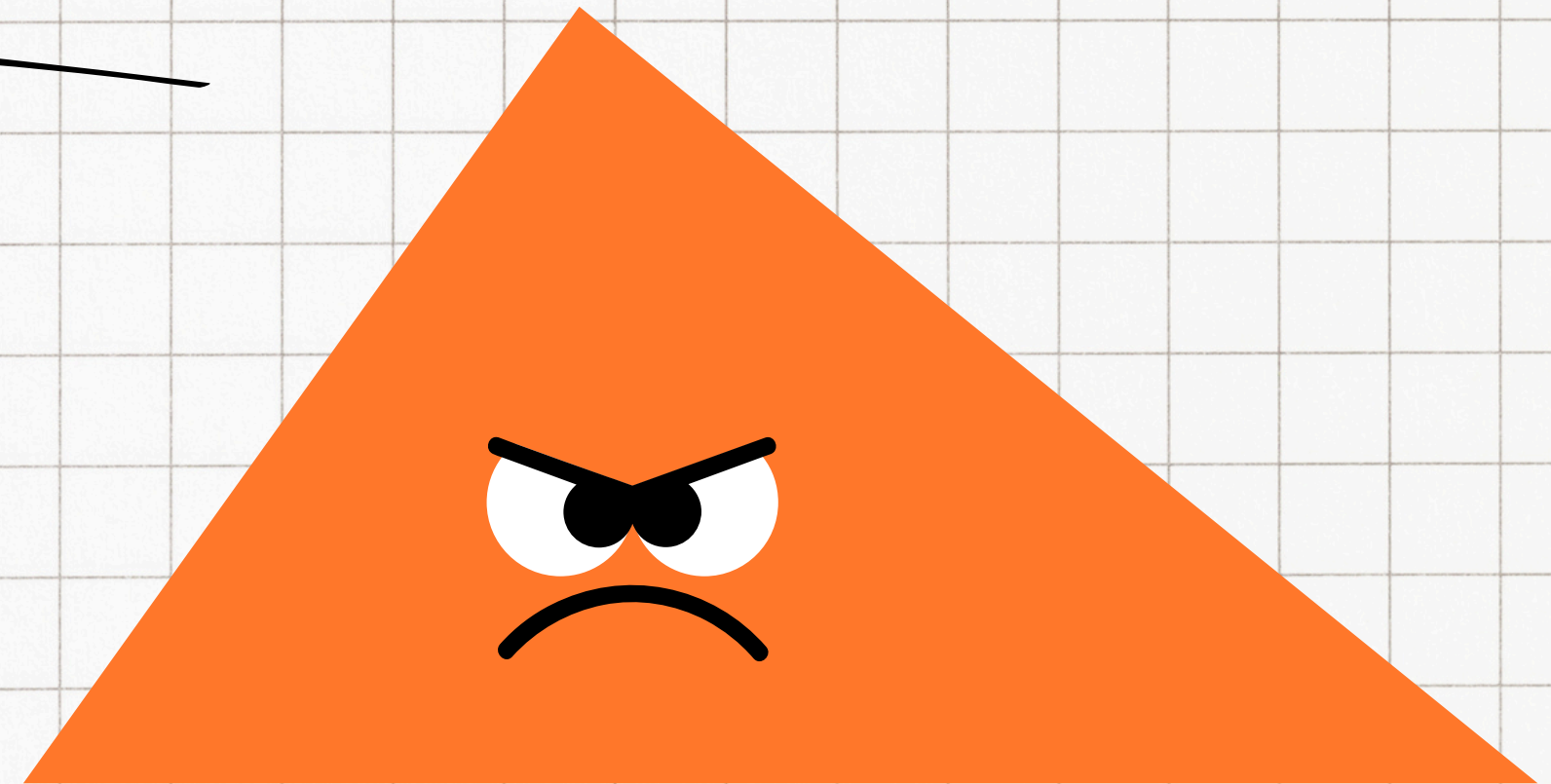


SCALED TRIANGLE



Definition: All three sides are of different lengths.

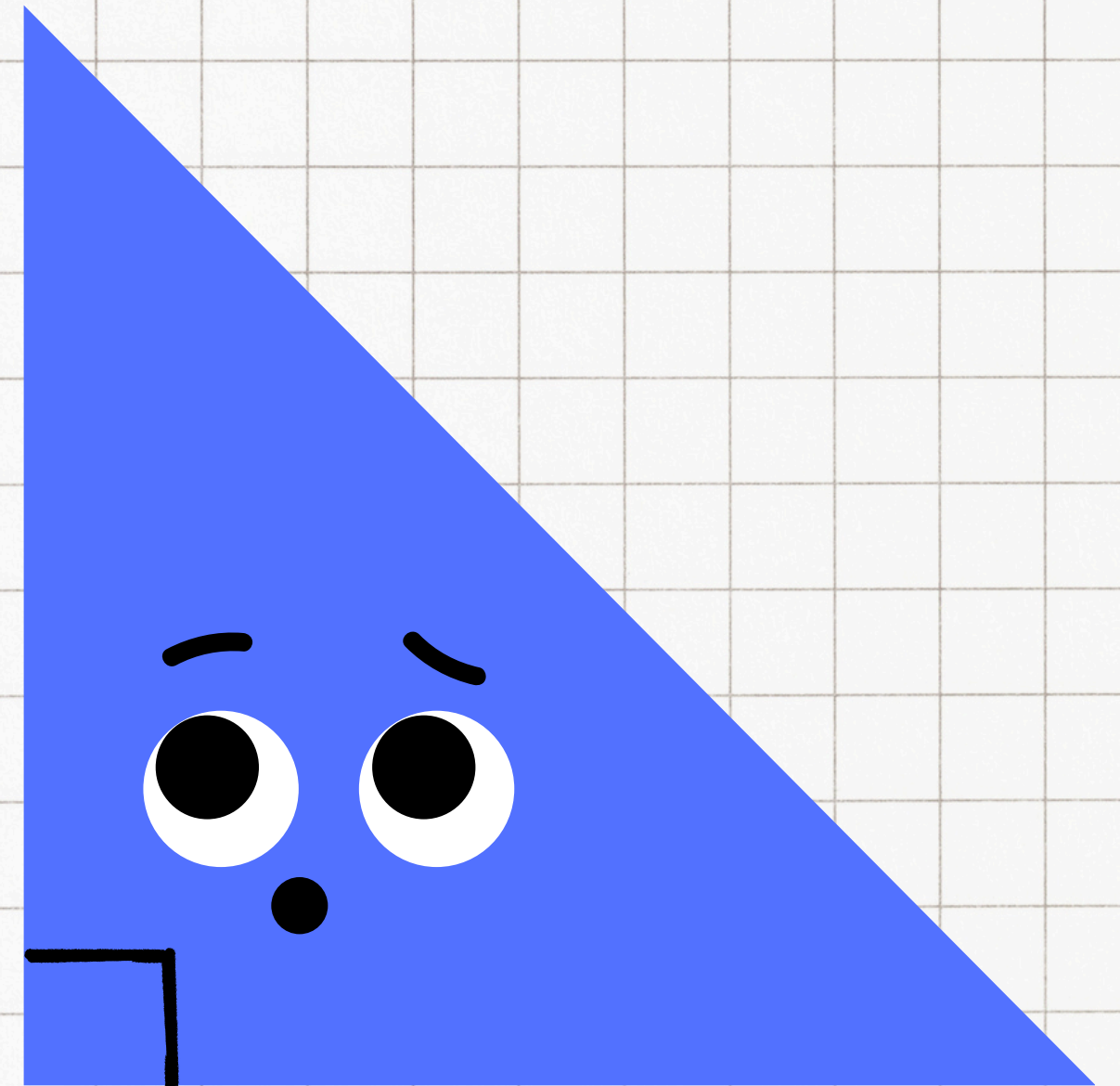
Properties: All three angles are different.



RIGHT TRIANGLE

Definition: One of the angles is exactly 90° .

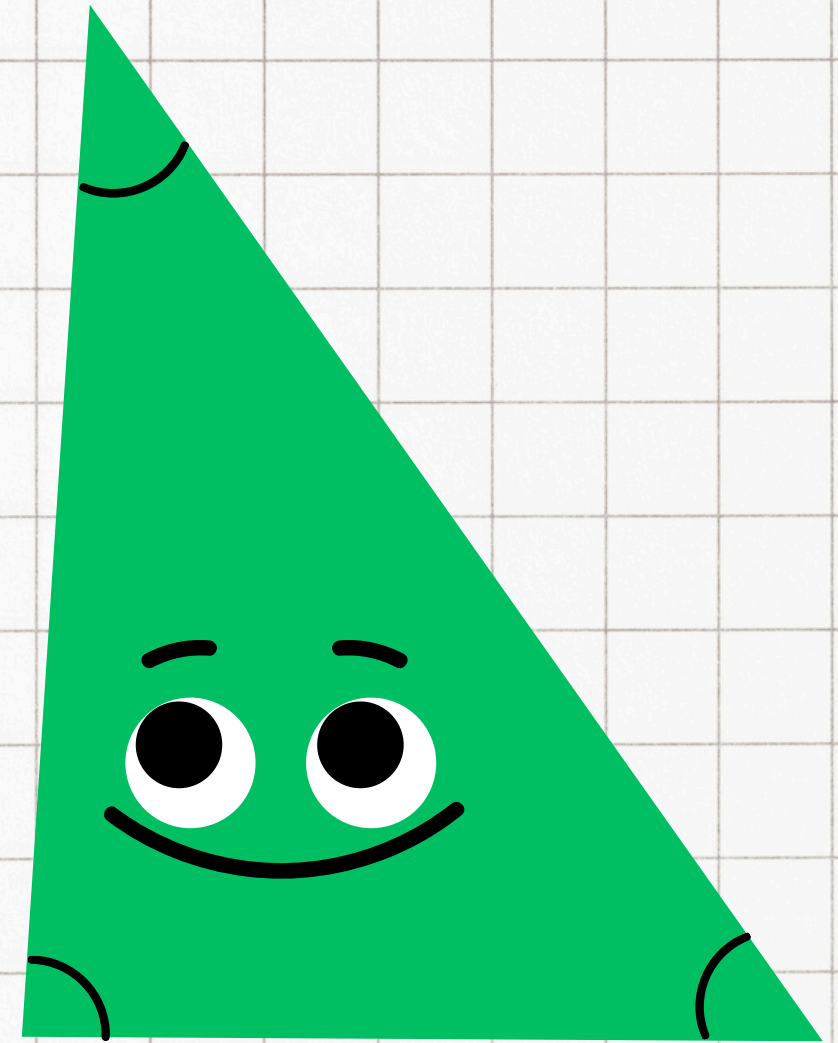
Properties: The side opposite the right angle is the hypotenuse, the longest side of the triangle.



ACUTE TRIANGLE

Definition: All three angles are less than 90° .

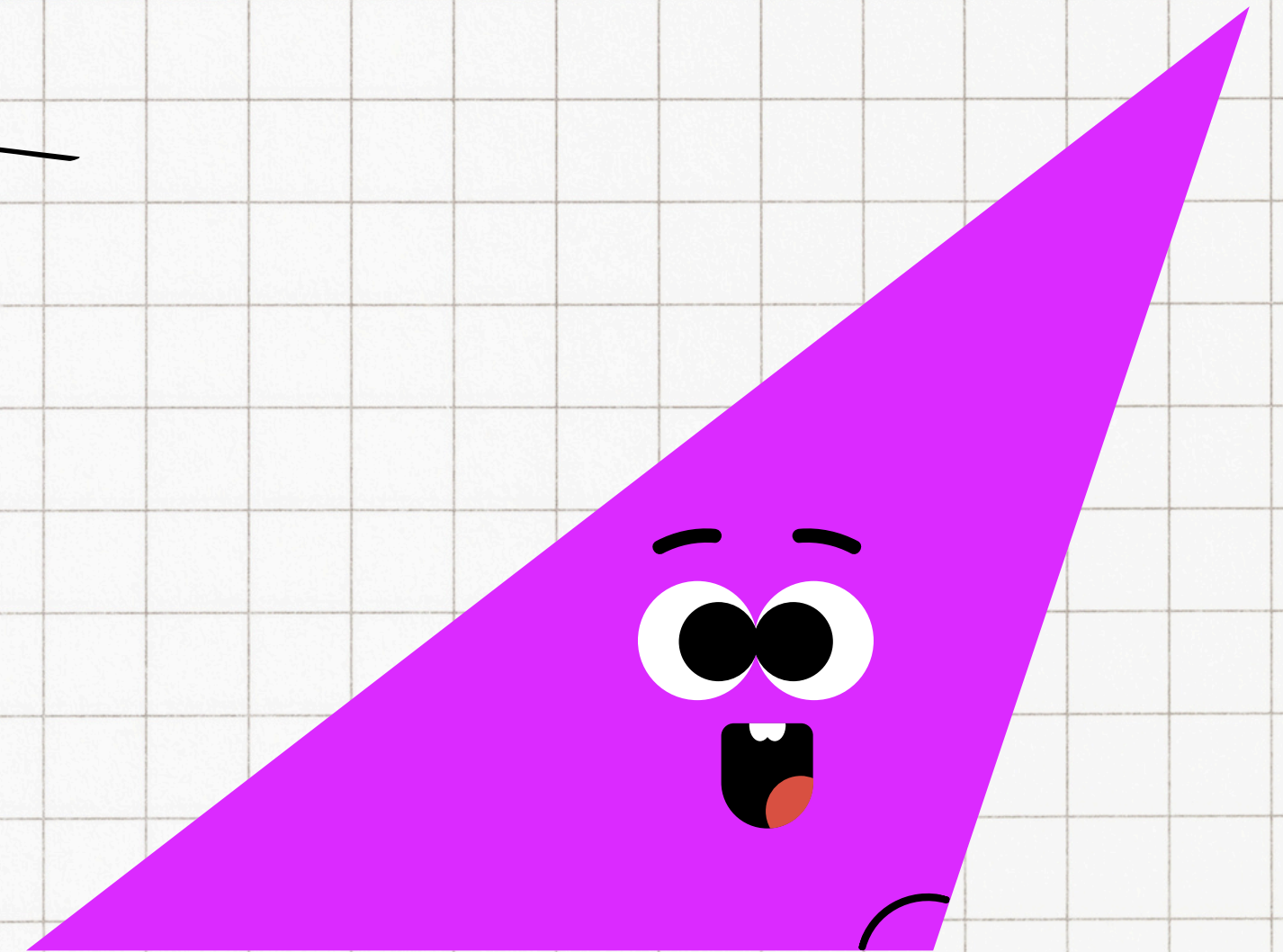
Properties: The triangle is sharp-angled, and the sum of the angles is still 180° .



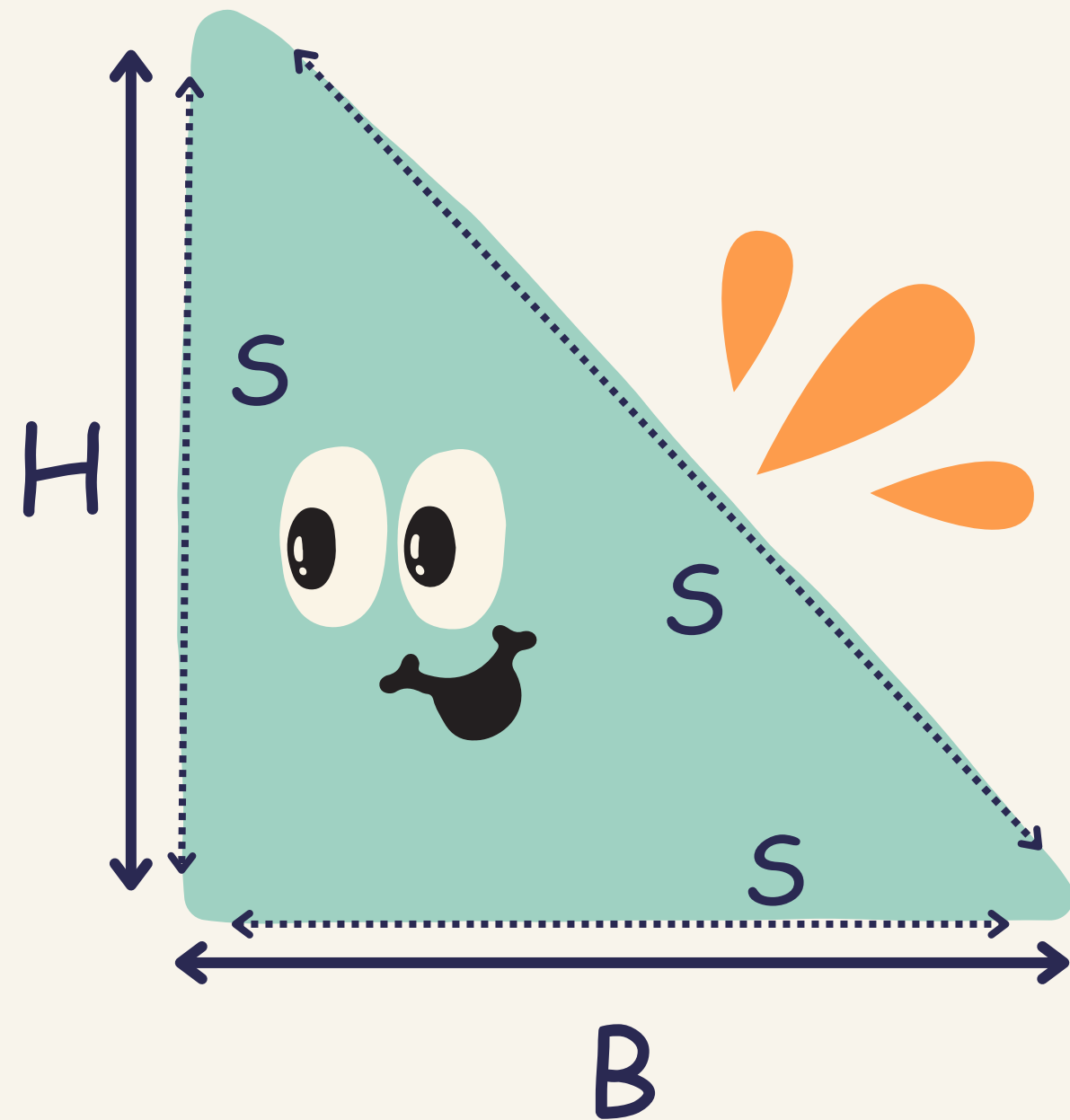
OBTUSE TRIANGLE

Definition: One of the angles is greater than 90° .

Properties: The other two angles are acute (less than 90°).



TRIANGLE



AREA:

$$\frac{1}{2} B \times H$$

PERIMETER:

$$S + S + S$$

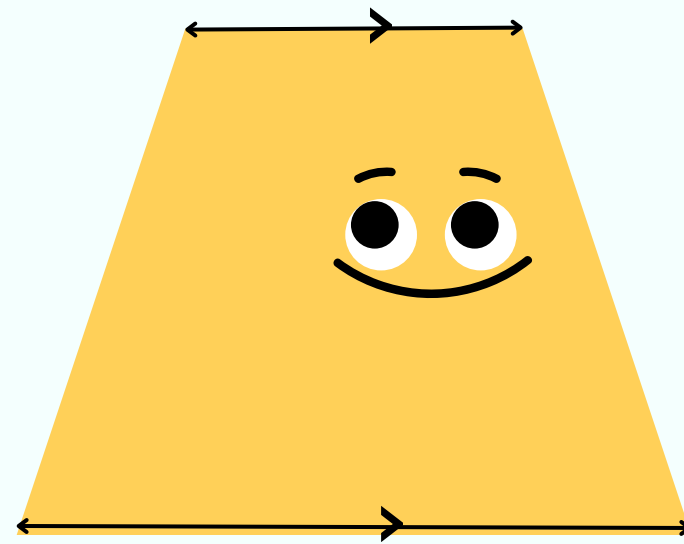
CLASSIFYING QUADRILATERALS

KITES



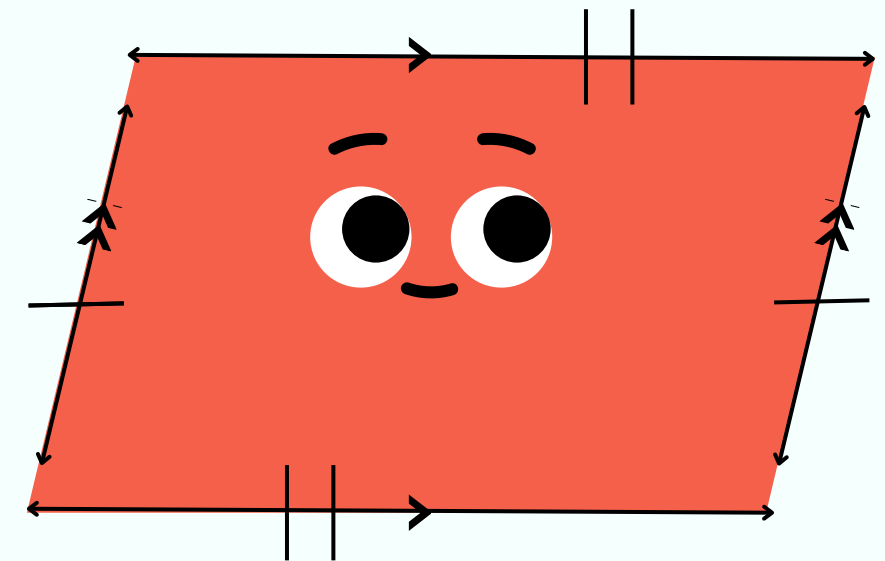
no parallel sides

TRAPEZOIDS



1 pair of parallel sides

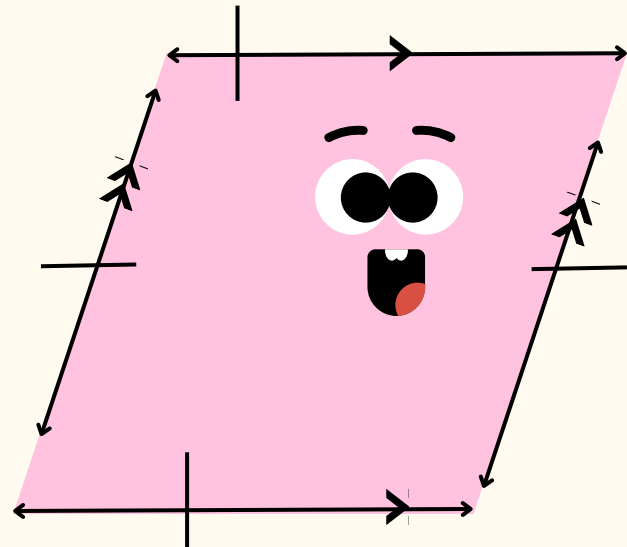
PARALLELOGRAMS



2 pairs of parallel sides

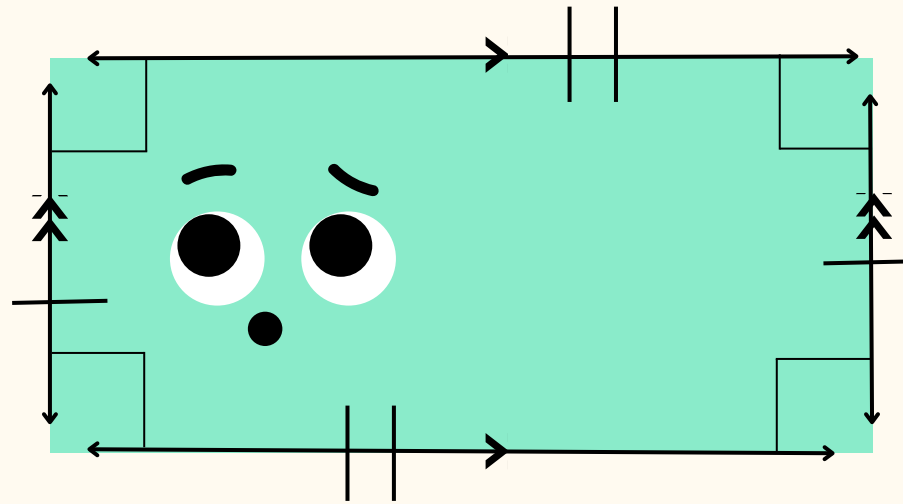
CLASSIFYING QUADRILATERALS

RHOMBUS



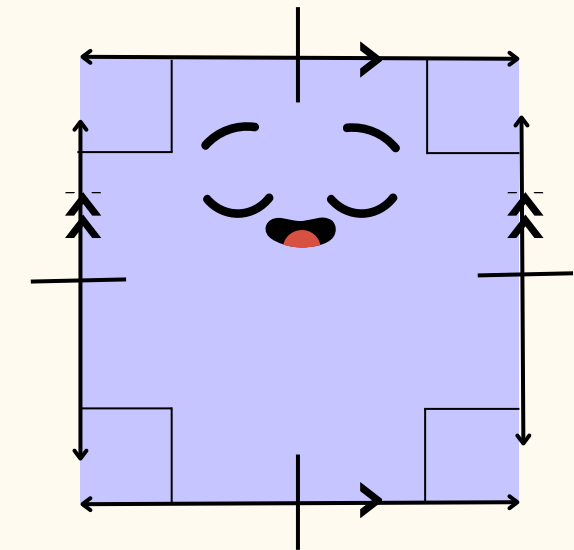
2 pairs of parallel sides
equal sides

RECTANGLES



2 pairs of parallel sides
4 right angles

SQUARES



2 pairs of parallel sides
4 right angles
equal sides

QUADRILATERALS

```
graph TD; A[QUADRILATERALS] --> B[KITE]; A --> C[PARALLELOGRAM]; A --> D[TRAPEZOID]; C --> E[RHOMBUS]; C --> F[RECTANGLE]; E --> G[SQUARE]; F --> G
```

KITE

PARALLELOGRAM

TRAPEZOID

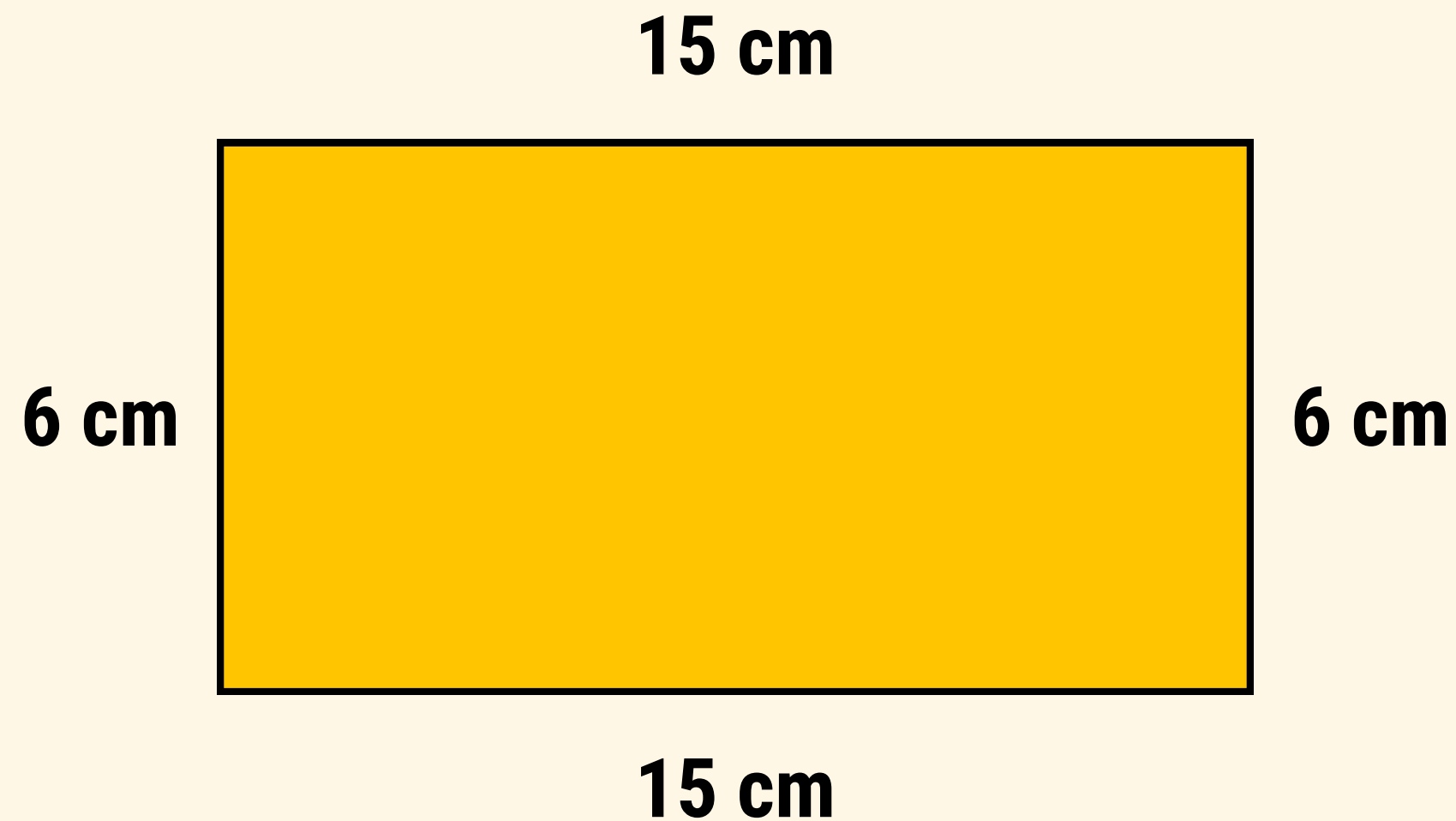
RHOMBUS

RECTANGLE

SQUARE

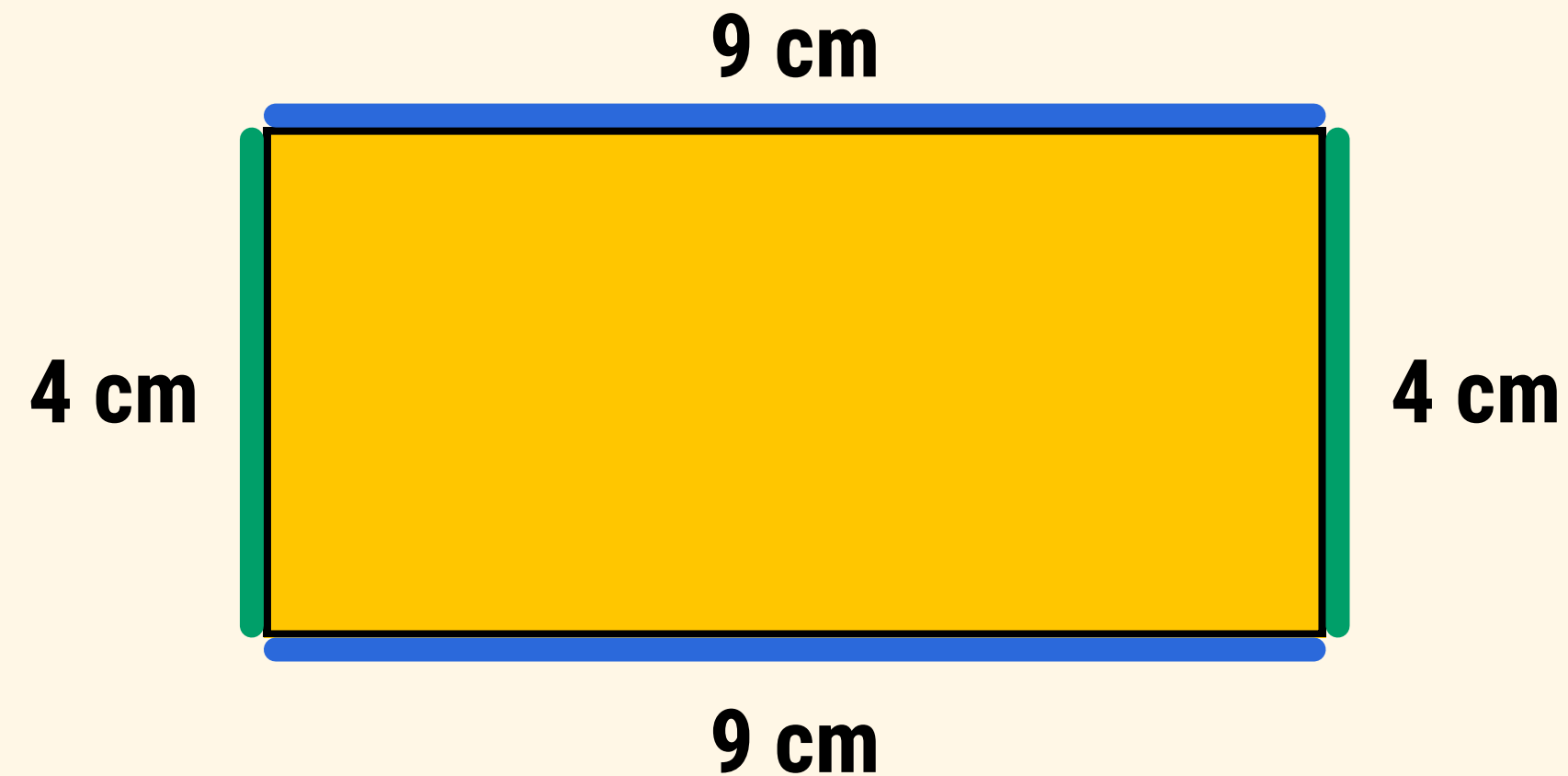
What is a Rectangle?

A rectangle is a type of quadrilateral, a four sided shape.



Side Properties

All sides are straight.
Opposite sides are equal in length.

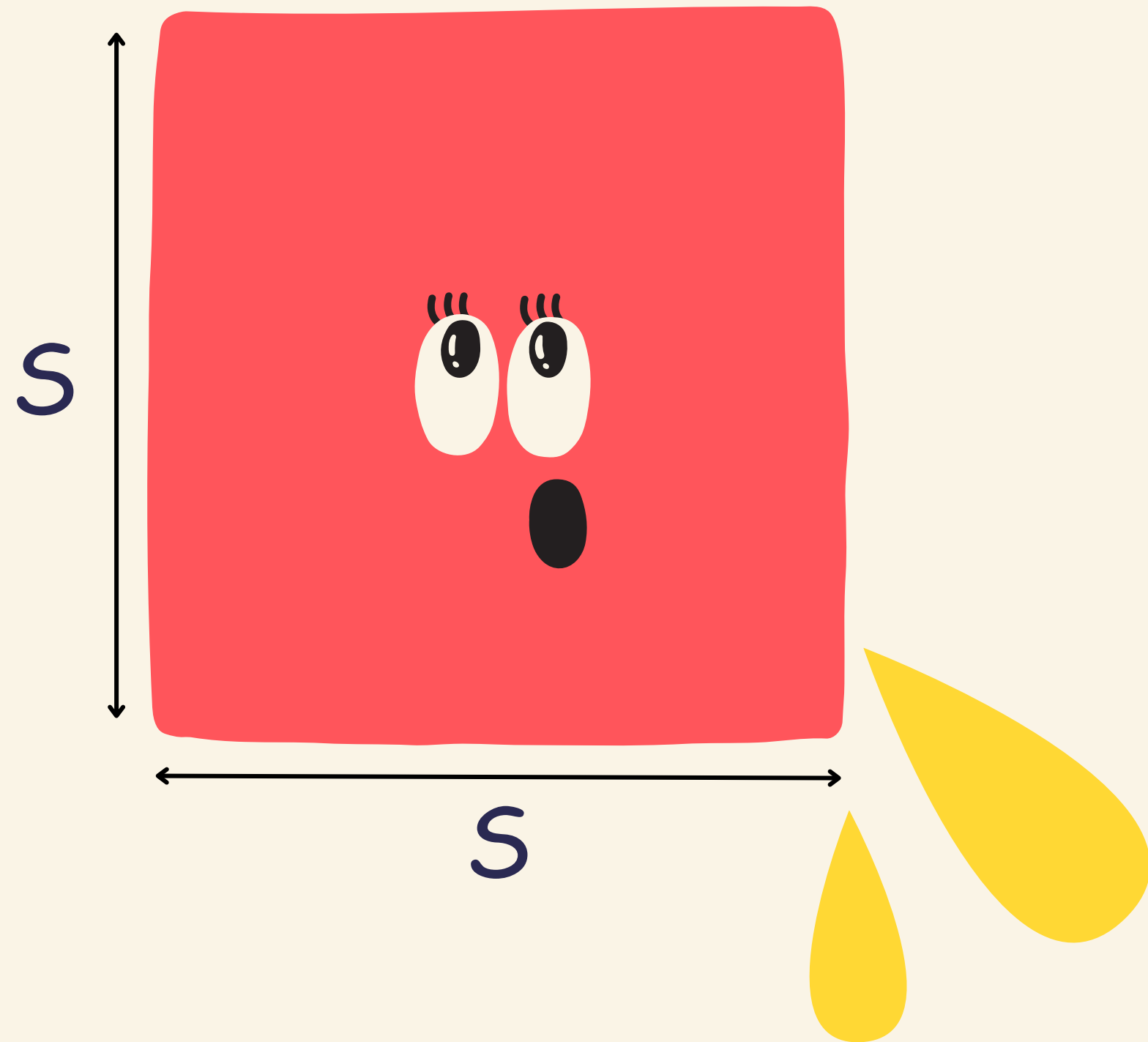


Angle Properties



All angles in a rectangle are right angles (90°).

S Q U A R E



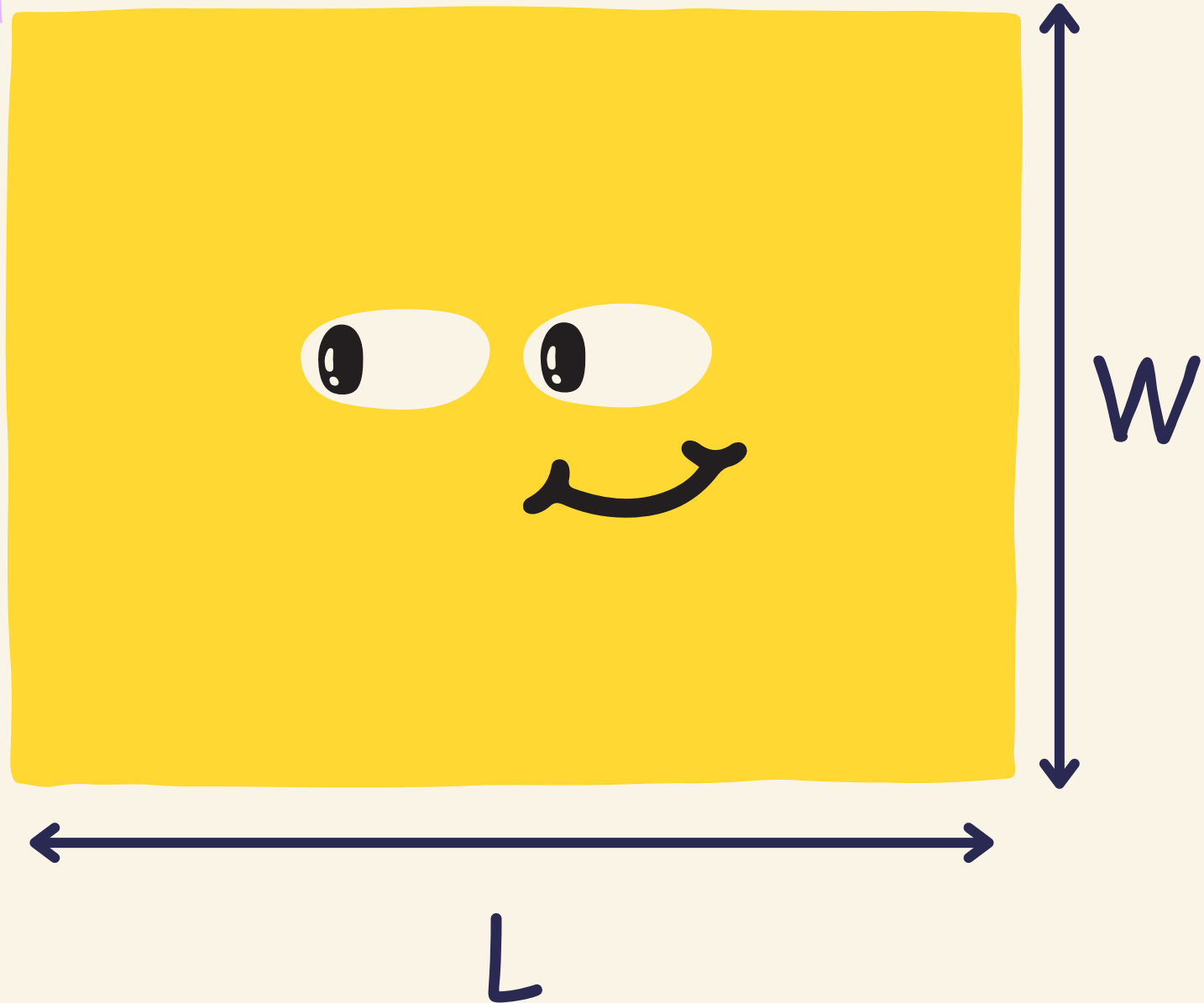
AREA:

$$S \times S$$

PERIMETER:

$$S + S + S + S$$

RECTANGLE



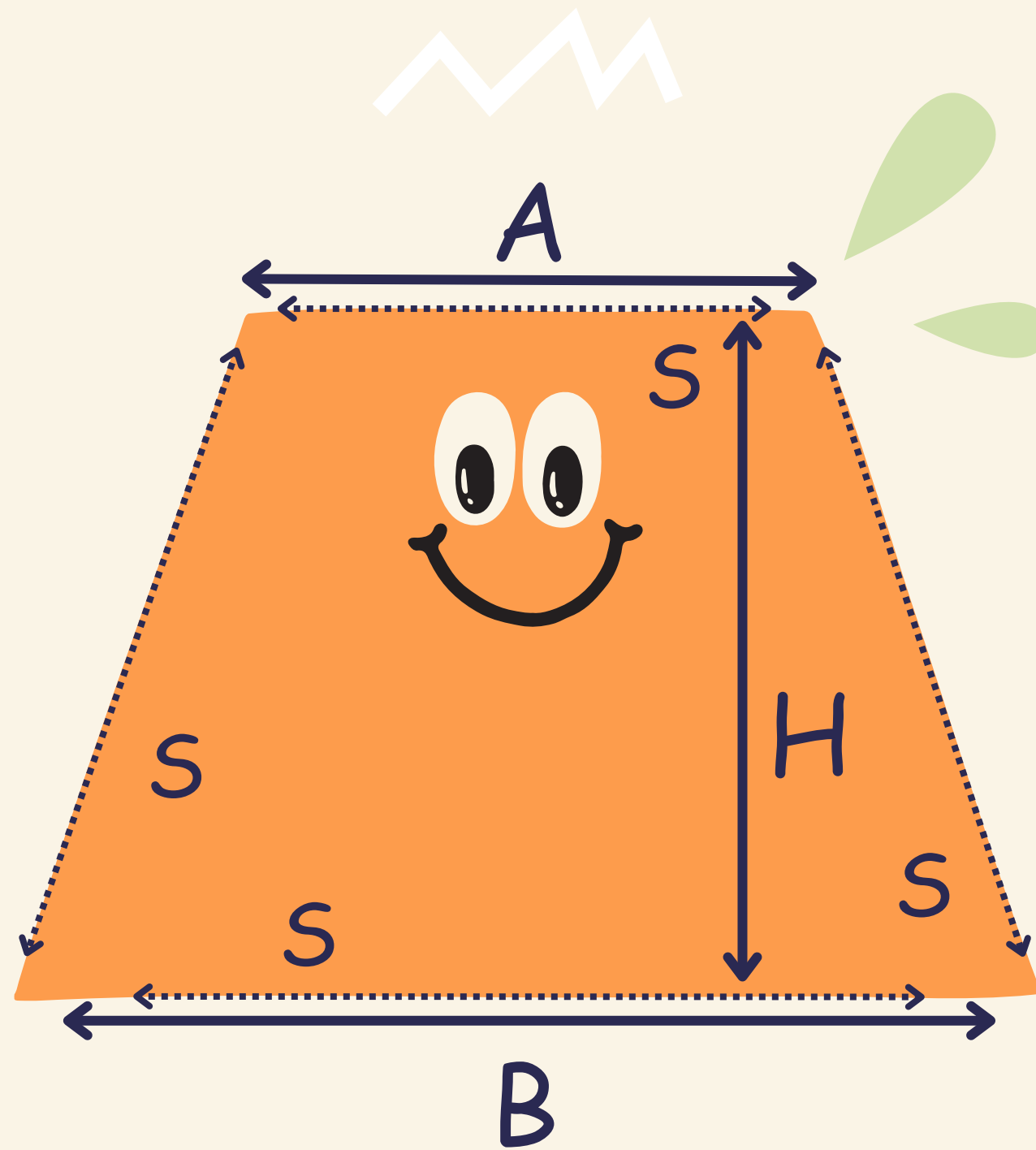
AREA:

$$L \times W$$

PERIMETER:

$$(L + W) \times 2$$

TRAPEZOID



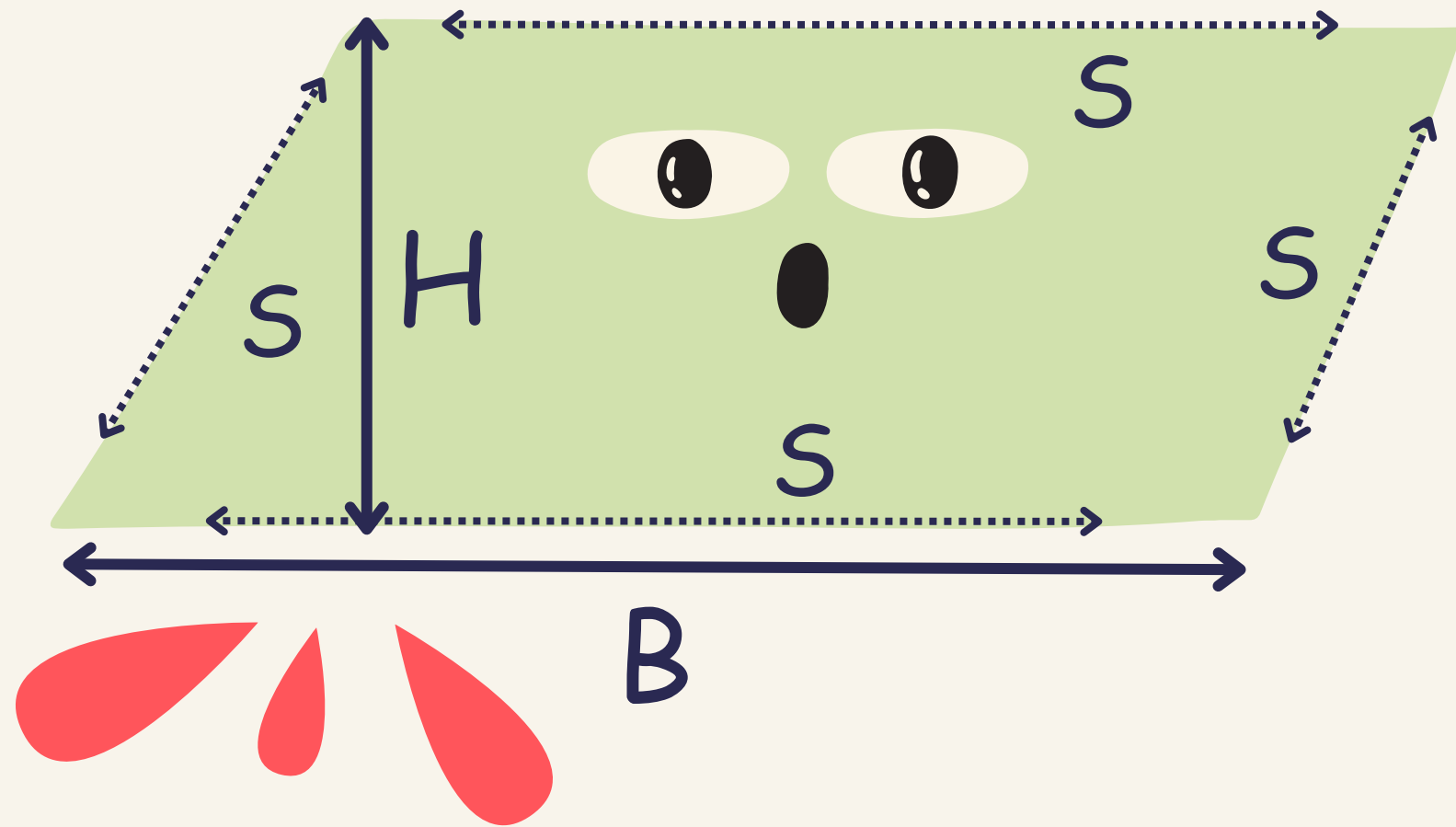
AREA:

$$\frac{1}{2} (A + B) \times H$$

PERIMETER:

$$S + S + S + S$$

RHOMBUS



AREA:

$$B \times H$$

PERIMETER:

$$S + S + S + S$$



THANK
YOU